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COHORT	**CYBERSHUJAA COHORT 1-2026**
TASK	**LINUX FUNDAMENTALS MODULE REPORT**

Contents

LINUX FUNDAMENTALS REPORT

Introduction

This report documents what I learned during the Hack The Box Linux Fundamentals module as part of the Cyber Shujaa program. The module introduced me to the Linux operating system and its core components, with a focus on practical command-line usage and system navigation relevant to cybersecurity.

Here is a shareable link to my completed module:

https://academy.hackthebox.com/achievement/2380391/18

1.0 Introduction

1.1 Linux Structure

In this module, I learned what Linux is, its history, and why it is important in cybersecurity. I understood that Linux is an open-source operating system with many distributions, and that the course uses Parrot OS through Pwnbox.

I learned the core Linux philosophy, which focuses on simplicity, modular tools, and storing configuration in text files. I also understood the main Linux components, including the kernel, shell, daemons, and system utilities, as well as how Linux is structured in layers.

Finally, I learned the Linux file system hierarchy and the purpose of key directories such as **/bin**, **/etc**, **/home**, and **/var**, which are essential for navigating and managing a Linux system.

1.2 Linux Distributions

In this module, I learned that Linux distributions are operating systems built on the Linux kernel, each designed for different use cases such as desktops, servers, and cybersecurity. I understood that while all distros share the same core structure and philosophy, they differ in tools, packages, and user experience.

I learned that Ubuntu and Fedora are popular for beginners and desktop use, Debian and CentOS are commonly used for servers, and Kali Linux and Parrot OS are widely used in cybersecurity due to their security-focused tools.

I also learned that Debian is known for its stability, reliability, and strong security support, making it suitable for long-term and enterprise use despite having a steeper learning curve.

1.3 Introduction to Shell

In this module, I learned what the Linux shell is and why it is essential, especially since many servers run on Linux. I understood that the shell is a text-based interface that allows direct interaction with the operating system through commands.

I learned the difference between the shell and the terminal, and how terminal emulators allow me to use the shell within a graphical environment. I was also introduced to tools like terminal multiplexers, which make it easier to work with multiple sessions at once.

Finally, I learned that Bash is the most commonly used Linux shell, and that using the shell allows for faster system control, automation, and scripting compared to relying only on the GUI.

2.0 The Shell

2.1 Prompt Description

In this module, I learned how to read and understand the Bash prompt. I understood that the prompt shows useful information such as the username, hostname, and current working directory, and that ~ represents the home directory. I also learned the difference between a normal user prompt ($) and a root prompt (#).

I learned that the appearance of the prompt is controlled by the PS1 variable and can be customized to display additional information like time, full directory paths, or other system details. This is especially useful during security tasks for better tracking and documentation.

2.2 Getting Help

In this module, I learned how to get help when using unfamiliar Linux commands. I learned how to use manual pages with the man command to view detailed documentation for tools and their options.

I also learned to use --help or -h flags to quickly view available command options, and the apropos command to search for tools related to a specific keyword. These methods help me understand and use new commands more efficiently when working in the terminal.

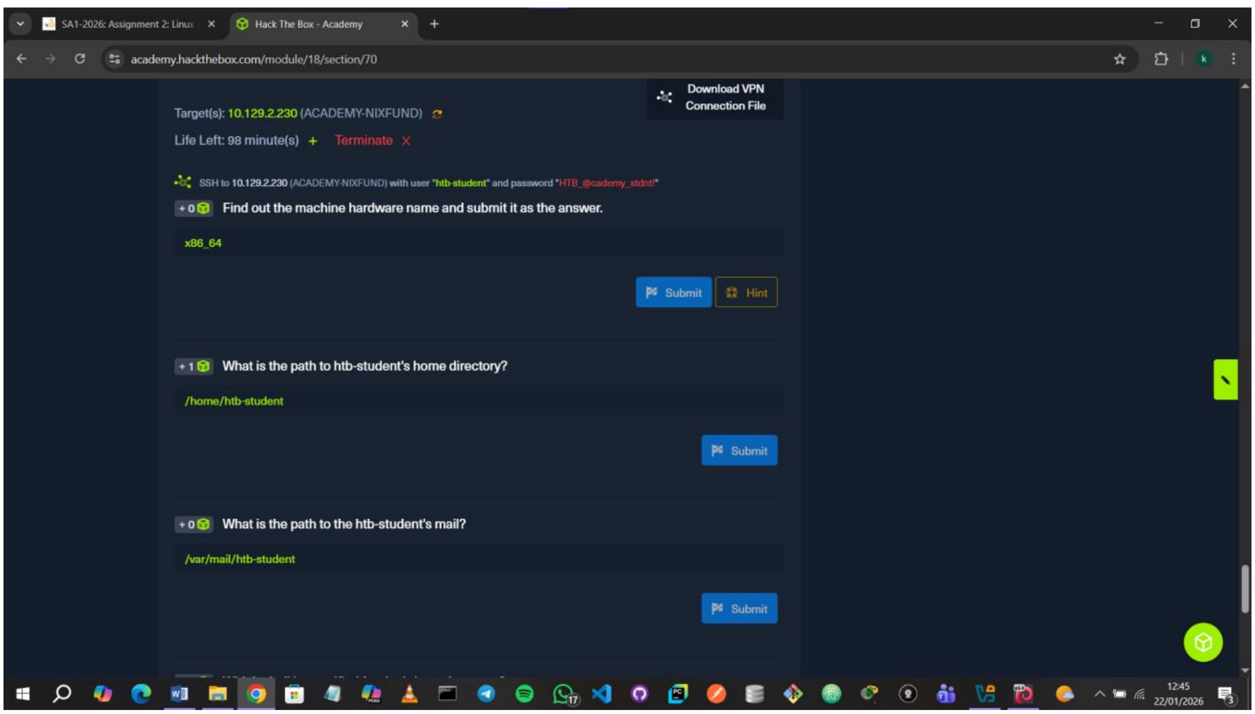
2.3 System Information

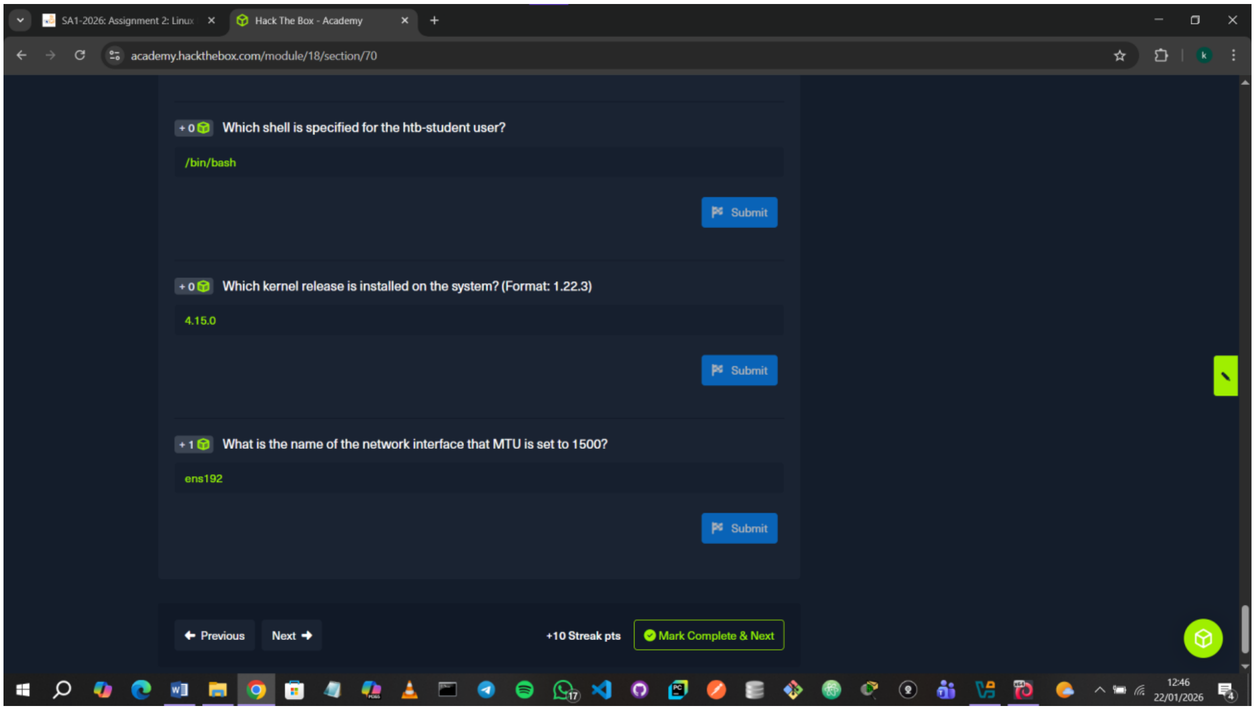
In this module, I learned how to gather important system information using common Linux commands. I practiced identifying the current user, hostname, system details, running processes, network information, and hardware devices using tools such as **whoami, id, hostname, uname**, and networking commands.

I also learned how to connect to remote Linux systems using SSH and understood why it is commonly used for remote administration and security testing. This knowledge is essential for understanding system configurations and identifying potential security issues.

For the questions I used such commands as: **uname –m** to find out the machine hardware name, **uname – a** to find out which kernel release is installed on the system and **ifconfig** to find the name of the network interface that MTU is set to 1500.


Home
File System
Trash
sf_Downloads
htb-student@nixfund: ~
Session Actions Edit View Help
Last login: Wed Sep 23 22:09:41 2020 from 10.10.14.6
htb-student@nixfund:~$ uname -m
x86_64
htb-student@nixfund:~$ pwd
/home/htb-student
htb-student@nixfund:~$ cd var
-bash: cd: var: No such file or directory
htb-student@nixfund:~$ cd /var/
htb-student@nixfund:/var$ ls
backups crash local log opt snap tmp
cache lib lock mail run spool www
htb-student@nixfund:/var$ cd mail/
htb-student@nixfund:/var/mail$ cd ~
htb-student@nixfund:~$ uname -a
Linux nixfund 4.15.0-123-generic #126-Ubuntu SMP Wed Oct 21 09:40:11 UTC 2020 x86_64 x86_64 x86_64 GNU/Linux
htb-student@nixfund:~$ ifconfig
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.129.6.99 netmask 255.255.0.0 broadcast 10.129.255.255
inet6 fe80::250:56ff:feb0:498b prefixlen 64 scopeid 0x20<link>
inet6 dead:beef::250:56ff:feb0:498b prefixlen 64 scopeid 0x0<global
>
ether 00:50:56:b0:49:8b txqueuelen 1000 (Ethernet)
RX packets 4877 bytes 341906 (341.9 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 732 bytes 62600 (62.6 KB)


SA1-2026: Assignment 2: Linux
Hack The Box - Academy
academy.hackthebox.com/module/18/section/70
Download VPN Connection File
Target(s): 10.129.2.230 (ACADEMY-NIXFUND)
Life Left: 98 minute(s) + Terminate X
SSH to 10.129.2.230 (ACADEMY-NIXFUND) with user "htb-student" and password "HTB_@cademy_stdnt!"
+ 0 Find out the machine hardware name and submit it as the answer.
x86_64
Submit
Hint
+ 1 What is the path to htb-student's home directory?
/home/htb-student
Submit
+ 0 What is the path to the htb-student's mail?
/var/mail/htb-student
Submit
12:45
22/01/2026



3.0 Workflow

3.1 Navigation

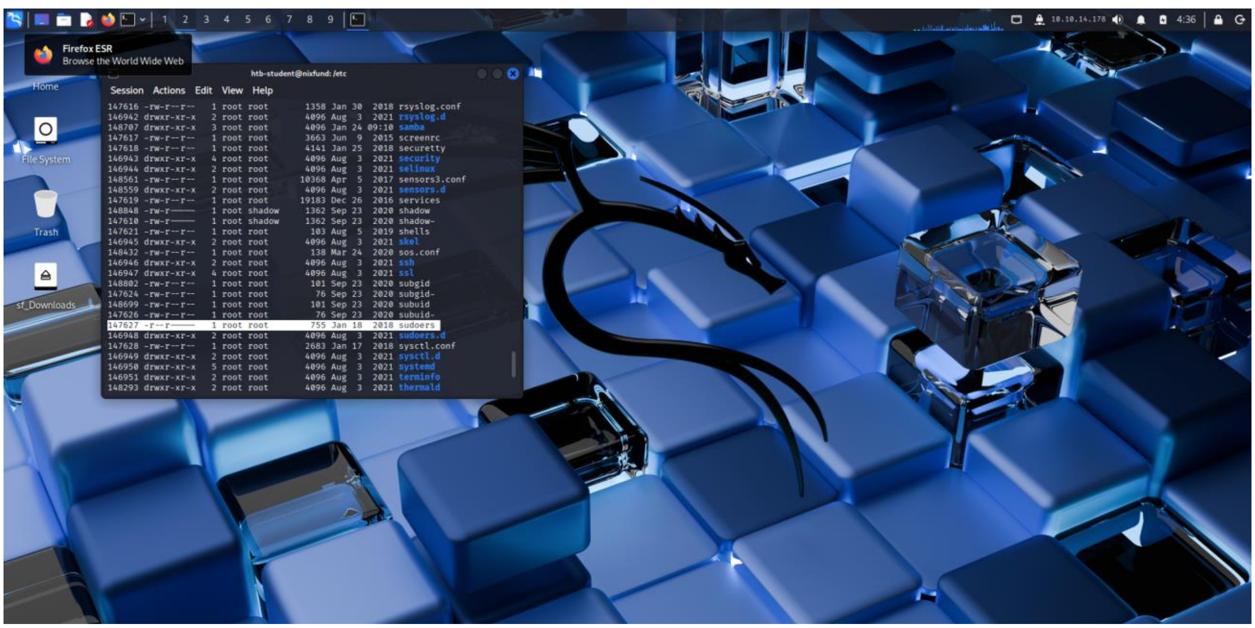
In this section, I learned how to navigate the Linux file system using the terminal. I practiced checking my current directory with pwd, listing files and folders using ls with different options, and viewing hidden files. I also learned how to move between directories using cd, use relative and absolute paths, and quickly return to previous directories.

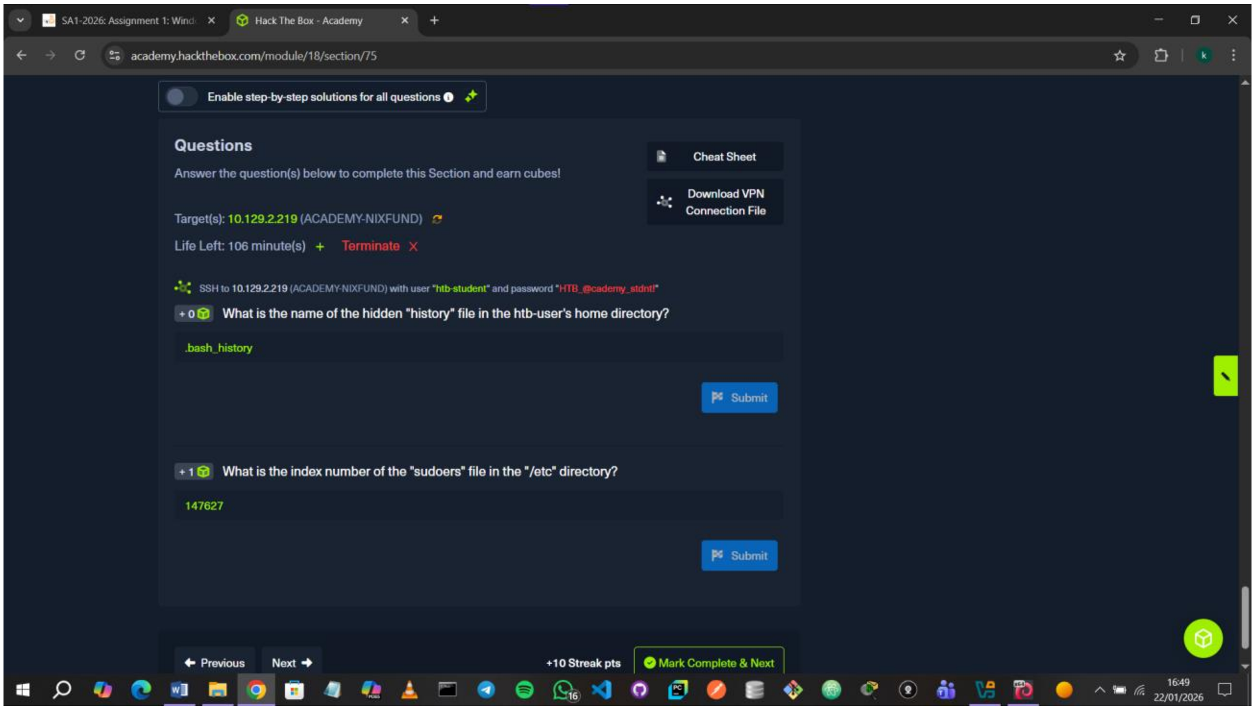
Question1: What is the name of the hidden history file in the htb-user's home directory?

Answer1: I found it by running **ls-la** and located it as the file with a single period in front of it.

Question2: What is the index number of the “sudoers” file in the “/etc” directory?

Answer2: I found it by going to the **/etc** directory by running **cd /etc/** and then running **ls –lai**.





3.2 Working with Files and Directories

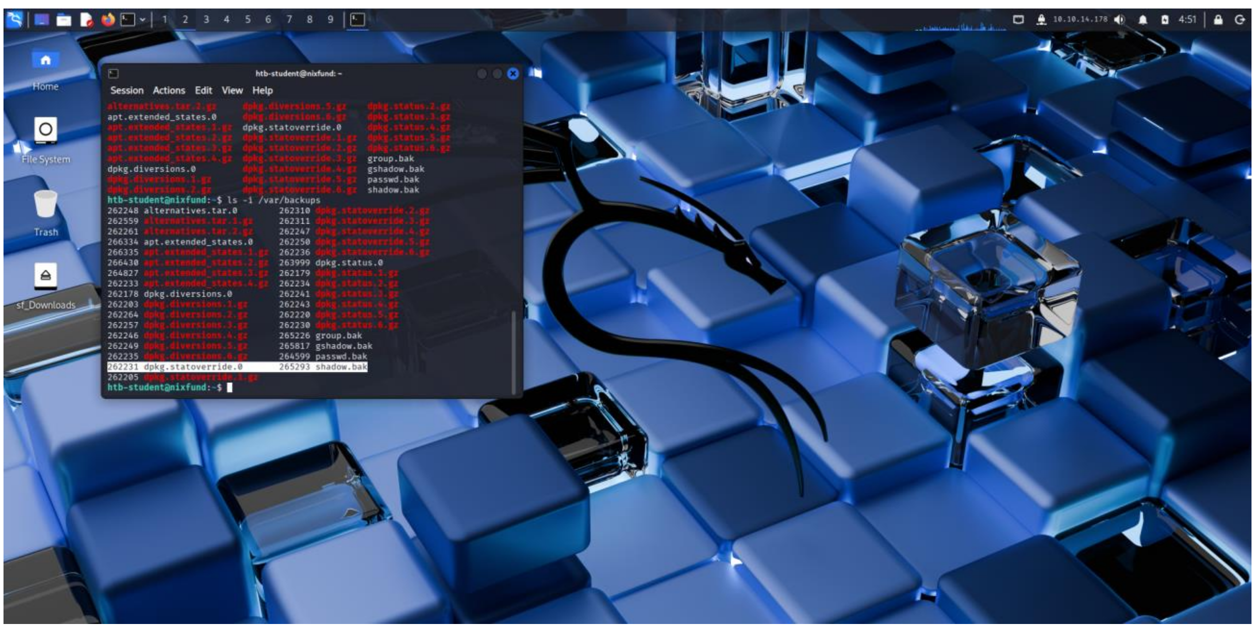
In this section, I learned how to create, organize, move, copy, rename, and delete files and directories using the Linux terminal. I used commands like touch, mkdir, mv, and cp, including creating nested directories with mkdir -p. I also learned how to work with file paths, verify structures using tree, and understand why the command line is faster and more flexible than using a graphical interface.

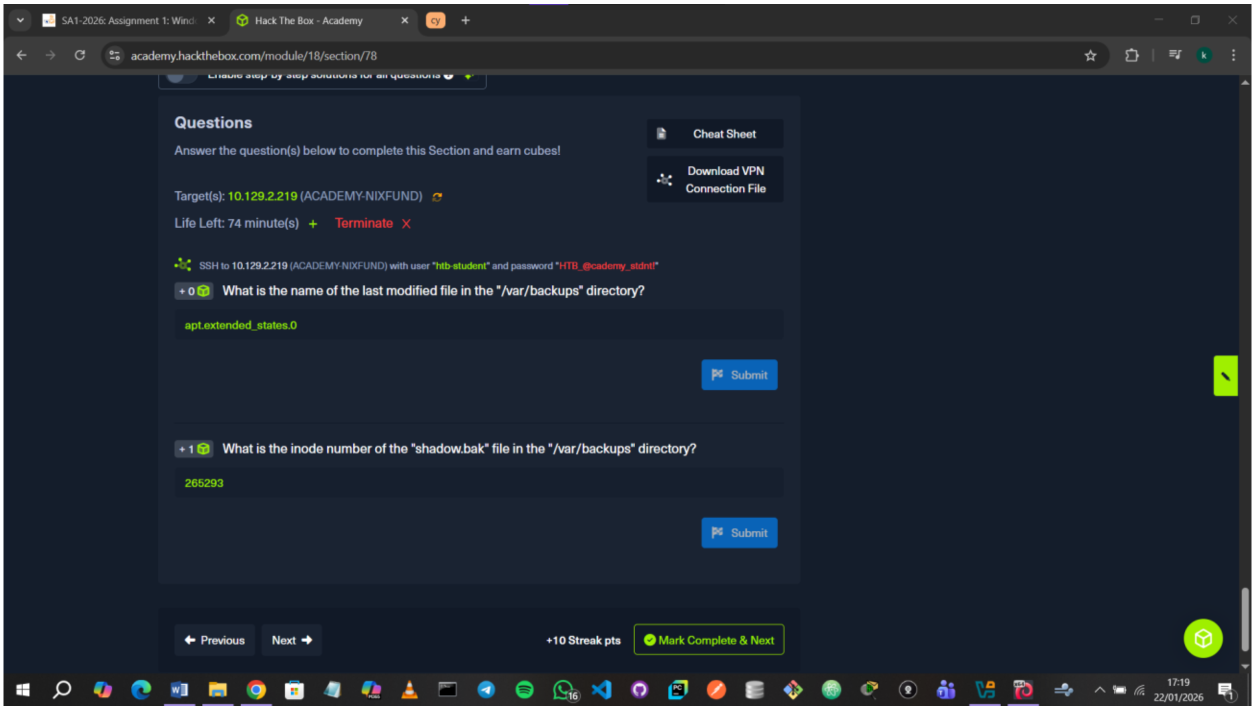
Question1: What is the name of the last modified file in the "/var/backups" directory?

Answer1: I listed all the files using **ls –la /var/backups** and located the file with the latest date.

Question2: What is the inode number of the “shadow.bak” file in the “/var/backups” directory?

Answer2: Had to google what the command was to find an inode number of a file. I ran **ls –i /var/backups** to find it.





3.3 Editing Files

In this section, I learned how to edit files in Linux using terminal-based text editors. I used **Nano** to quickly create, edit, search, save, and view files, and understood its basic shortcuts. I was also introduced to **Vim**, learned that it is a modal editor, and explored its different modes and the vimtutor tool for practice.

3.4 Find Files and Directories

I learned to use **which** to check if tools like Python are installed and where they're located. I also used **find** to search for files using filters like name, size, owner, and date, and **locate** for faster searches using a system database.

Question1: What is the name of the config file that has been created after 2020-03-03 and is smaller than 28k but larger than 25k?

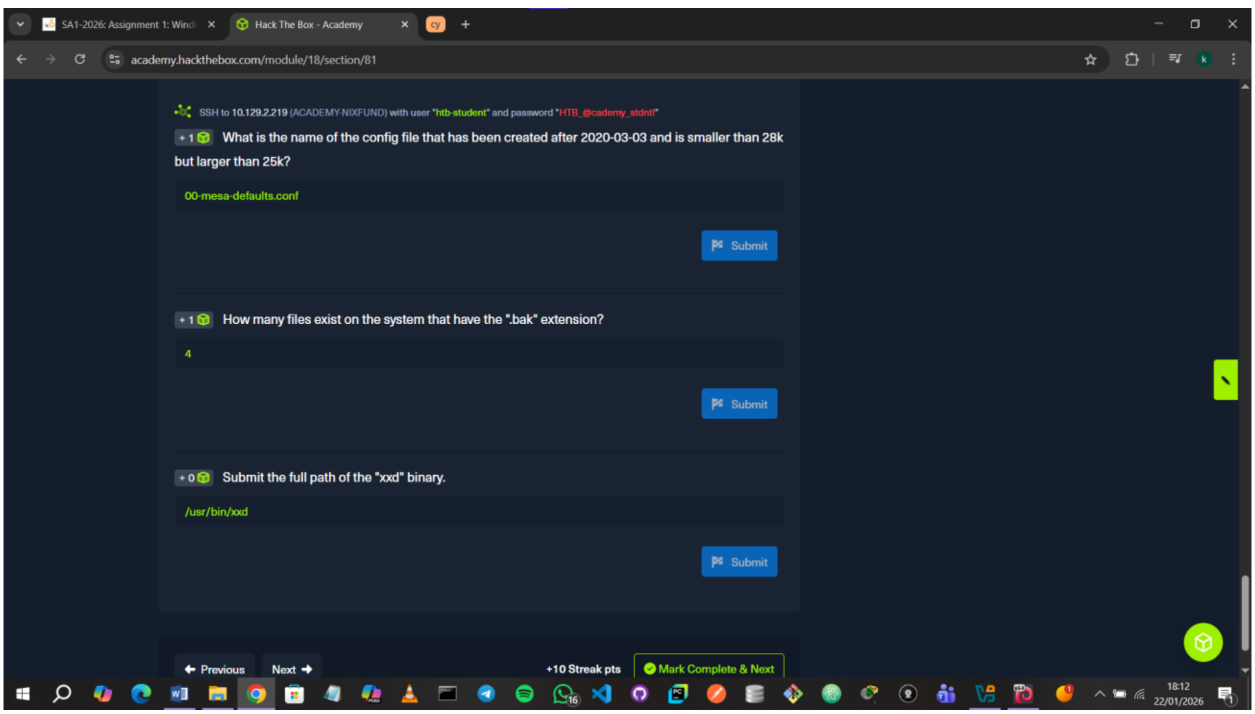
Answer1: I enlisted ChatGPT's help to help me solve this. I ran **find / -type f -name *.conf -size +25k -size -28k -newermt 2020-03-03 -exec ls -al {} \; 2>/dev/null** to find the name.

Question2: How many files exist on the system that have the ".bak" extension?

Answer: I found the number of files by running **find / -type f -name *.bak -exec ls -al {} \; 2>/dev/null**.

Question3: Submit the full path of the “xxd” binary.

Answer3: I found the path by running **which xxd**.



3.5 File Descriptors and Redirections

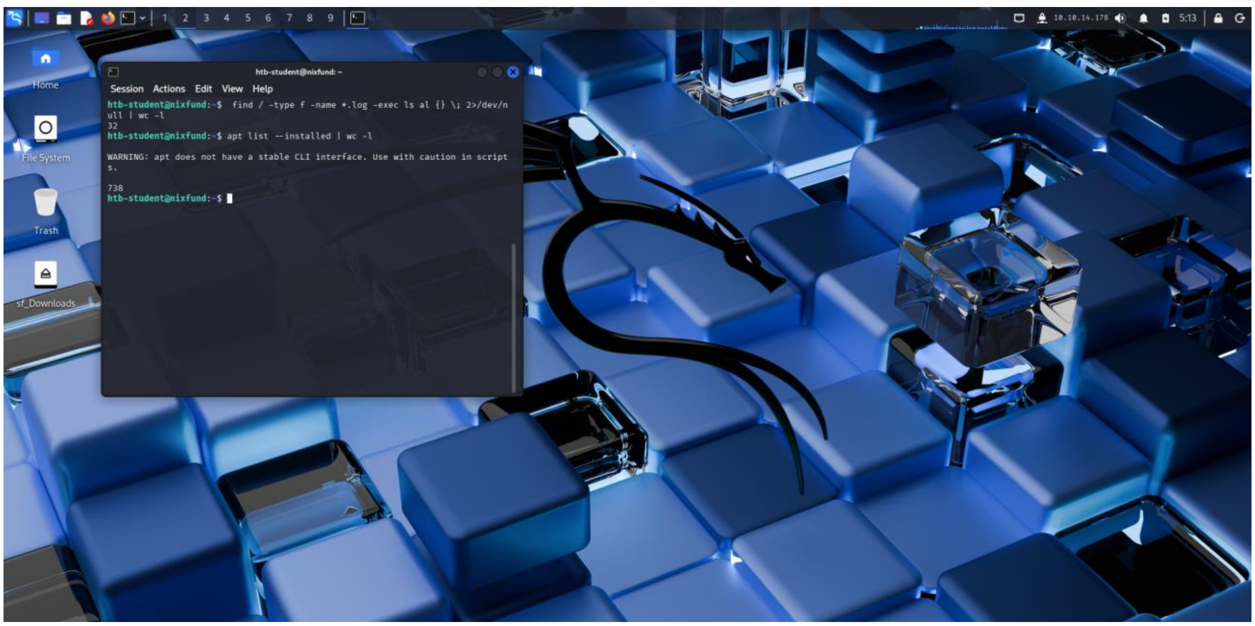
This part introduced file descriptors and redirections, which control how input, output, and errors flow between commands and files. I learned that Linux separates normal output, errors, and input, and that these streams can be redirected to files, ignored, or passed into other commands using pipes. I didn't fully grasp this section yet, but I understood that it's about controlling where command output goes.

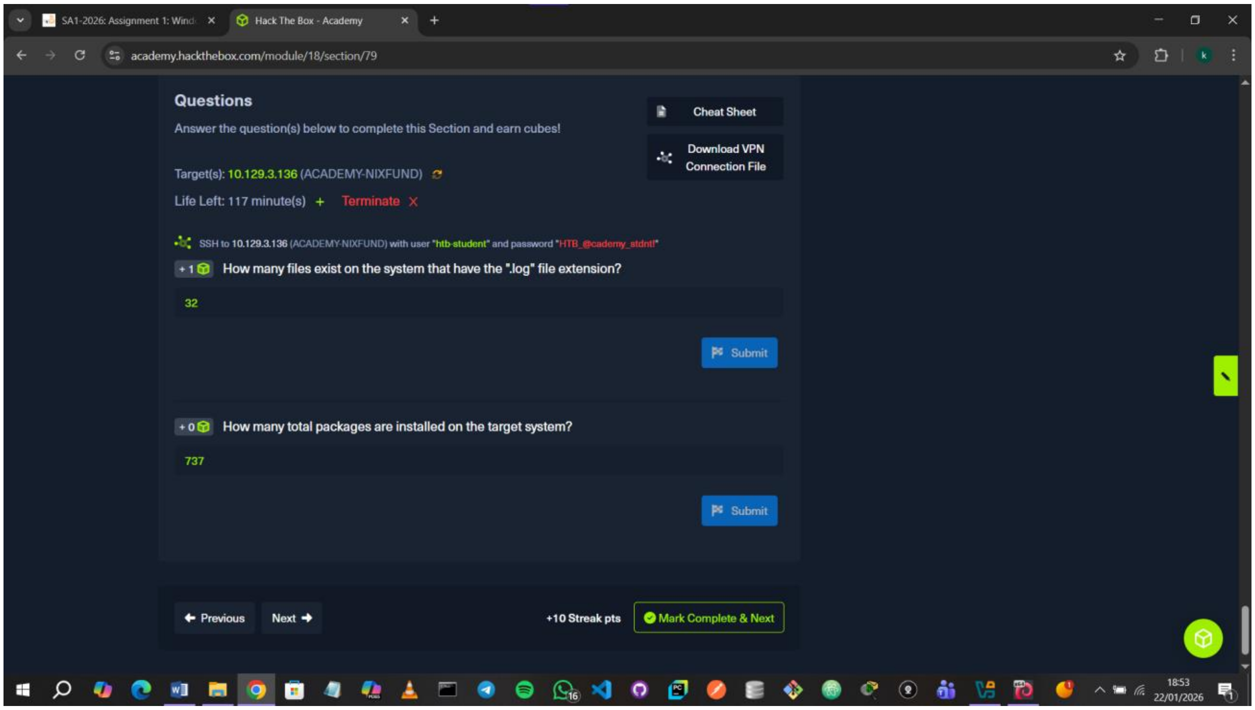
Question1: How many files exist on the system that have the ".log" file extension?

Answer1: Didn't understand this section of the module that well but I had done a similar-ish question in the previous section. I ran **find / -type f -name *.log -exec ls -al {} \; 2>/dev/null | wc –l**.

Question2: How many total packages are installed on the target system?

Answer2: Had to google how to get the number of packages installed on a Linux system. I ran **apt list –installed | wc -l** to get the number. I found 738 packages but for some reason HTB only accepted 737.



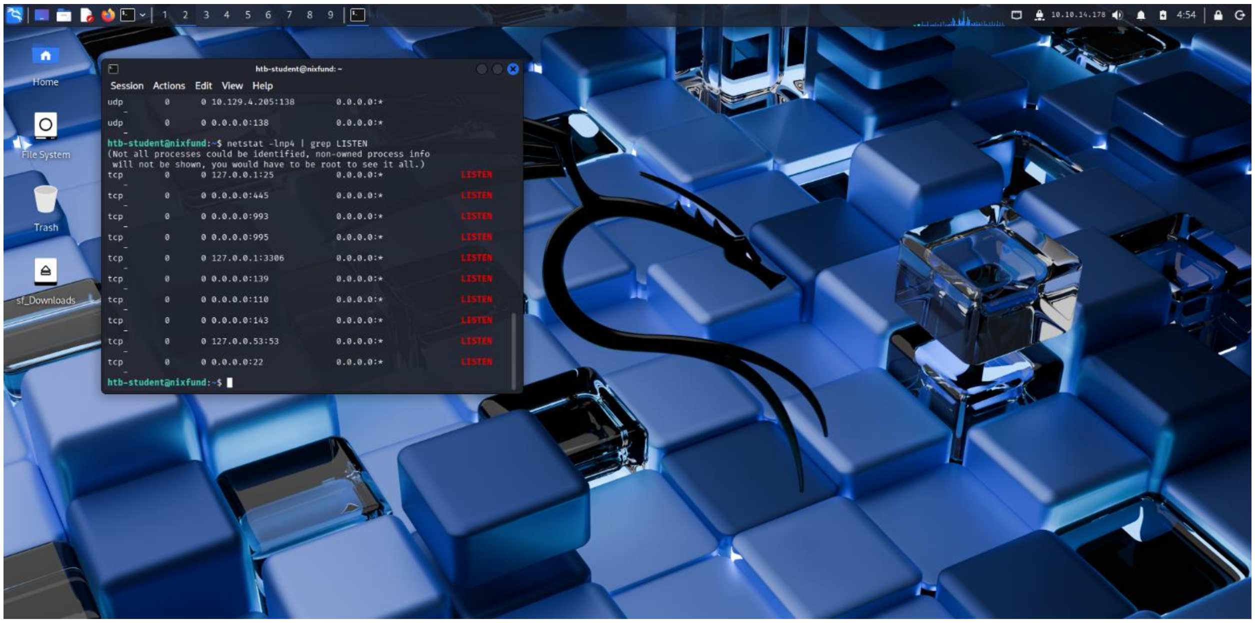


3.6 Filter Contents

I learned how to view and filter file contents directly from the terminal using tools like more, less, head, and tail. I also learned how to search, sort, and process text using commands such as grep, sort, cut, tr, awk, sed, and wc, and how to combine them with pipes to extract only the information I need.

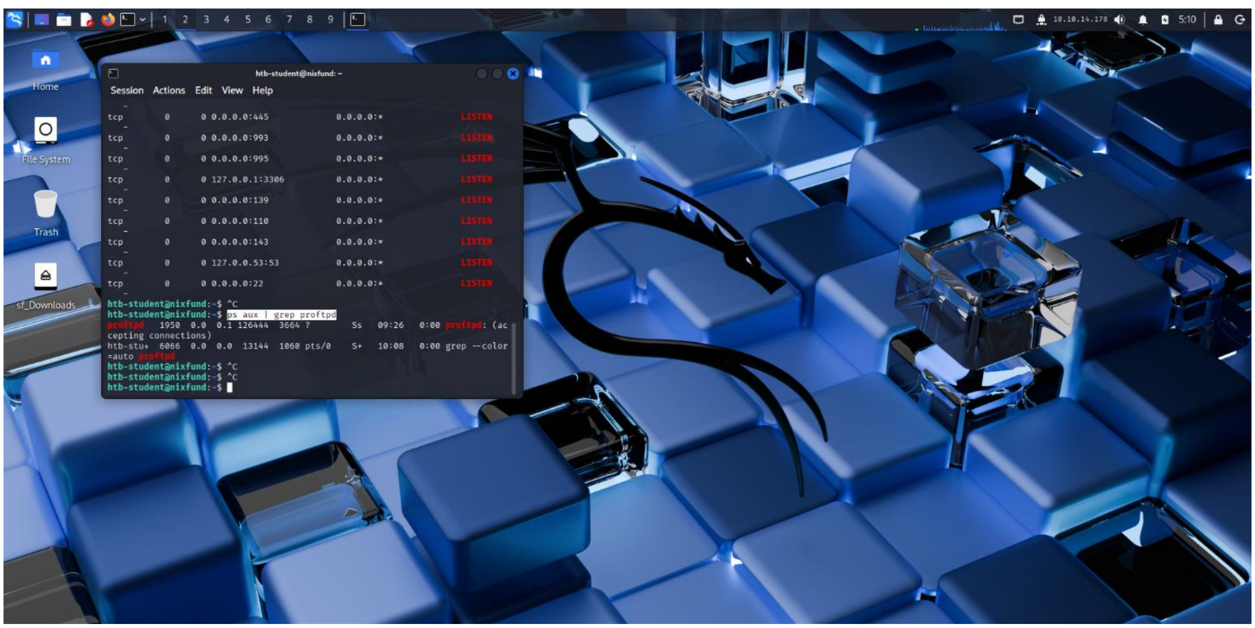
Question1: How many services are listening on the target system of all interfaces? (Not on local host and IPv4 only)

Answer1: I ran **netstat -lnp4 | grep LISTEN** using grep to filter out services that were not listening and counted those listening and not on local host as 7.



Question2: Determine what user the ProFTPd server is running under. Submit the username as the answer.

Answer2: Googled the command for this as **ps aux | grep proftpd**. It was difficult figuring out the username. Eventually found out it was **proftpd**.



Question3: Use cURL from your Pwnbox (not the target machine) to obtain the source code of the "https://www.inlanefreight.com" website and filter all unique paths (https://www.inlanefreight.com/directory" or "/another/directory") of that domain. Submit the number of these paths as the answer.

Answer3: I ran **curl https://www.inlanefreight.com | grep inlane | tr " " "\n" | sort -u | grep -E 'src|href'** using grep to only f show what I needed.

I then used **wc –l** to display the count which was 37.

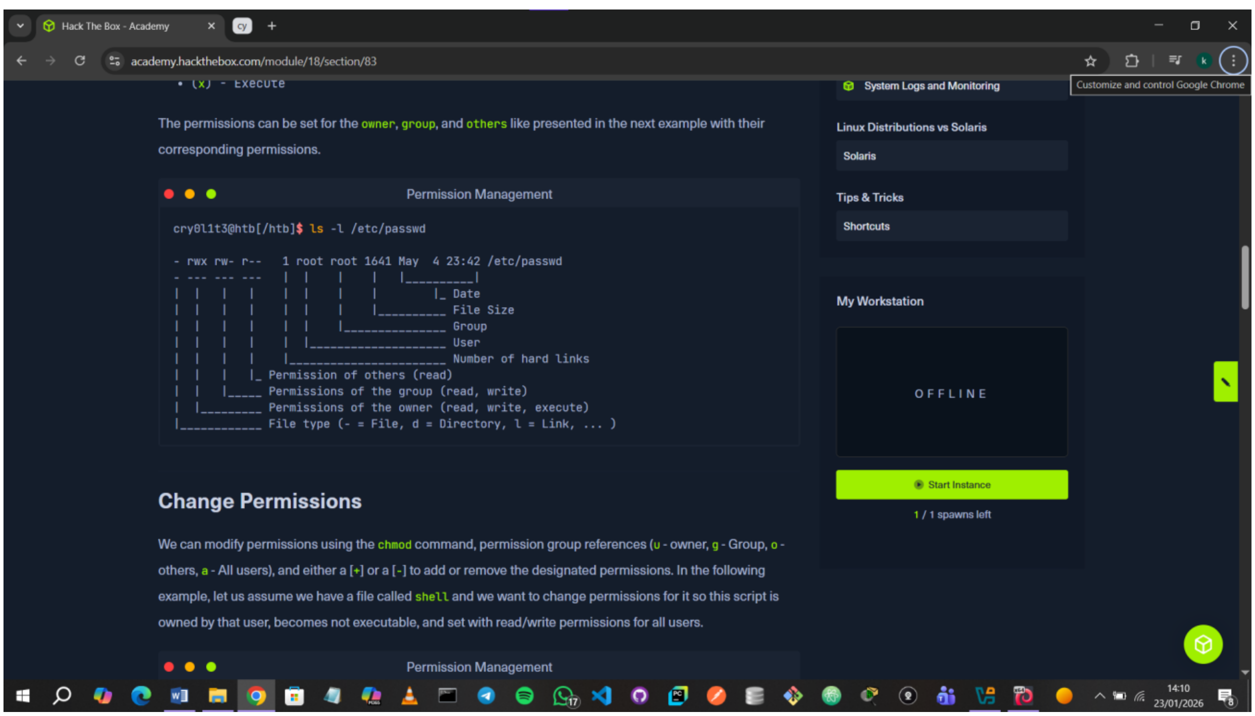
3.7 Regular Expressions

I learned how to use regular expressions to search and filter text using patterns instead of exact words, which makes it easier to extract specific information from files when working with tools like grep

3.8 Permission Management

I learned that Linux permissions act like keys that control who can access or modify files and directories. Every file has an owner and a group, and permissions are set for the owner, the group, and others. Read, write, and execute are the three basic permissions, and these can be combined and represented using letters or octal numbers.

I also learned how to change permissions with chmod and change ownership with chown. Special permissions like SUID and SGID let users run files with the privileges of the file's owner or group, which can be powerful but risky. The sticky bit adds extra security to shared directories, allowing only file owners or administrators to delete or rename files.



4.0 System Management

4.1 User Management

I learned that managing users in Linux is all about controlling who can do what on a system. I practiced creating users, assigning them to groups, and modifying their accounts using commands like useradd, usermod, and passwd. I also explored how to execute commands as another user with sudo or su, which is crucial for performing administrative tasks safely. Understanding user accounts, groups, and permissions helps me maintain system security, manage access, and troubleshoot issues effectively.

Question1: Which option needs to be set to create a home directory for a new user using "useradd" command?

Answer1: I ran **man useradd**. The option was –m.

Question2: Which option needs to be set to lock a user account using the "usermod" command? (Long version of the option).

Answer2: I ran **man usermod**. The long version was –lock.

Question3: Which option needs to be set to execute a command as a different user using the “su” command? (Long version of the option)

Answer3: I ran **man su**. The answer was –command.

4.2 Package Management

I learned that Linux package management is essential for installing, updating, and removing software efficiently. Packages contain software binaries, configuration files, and dependency information, and package managers handle these automatically to make installation and maintenance easier.

4.3 Service and Process Management

I learned that Linux services, or daemons, run in the background to keep the system operational. System services start at boot, while user-installed services add extra features. I also learned about processes, their PIDs, and controlling them with commands like ps, kill, and signals such as SIGTERM or SIGKILL.

Question1: Use the "systemctl" command to list all units of services and submit the unit name with the description "Load AppArmor profiles managed internally by snapd" as the answer.

Answer1: I found it as "snapd.apparmor.service" by running **systemctl list-units --type=service | grep "Load AppArmor".**

4.4 Task Scheduling

I learned that task scheduling in Linux allows tasks to run automatically at specific times or intervals without manual input. I explored **systemd timers** and **cron** as the main scheduling tools. Systemd uses timers and services to trigger scripts based on time or events, while cron uses crontab entries with defined time fields. I also learned why task scheduling is important for system administration and cybersecurity, especially for detecting persistence mechanisms like malicious cron jobs.

Question1: What is the Type of the service of the "dconf.service"?

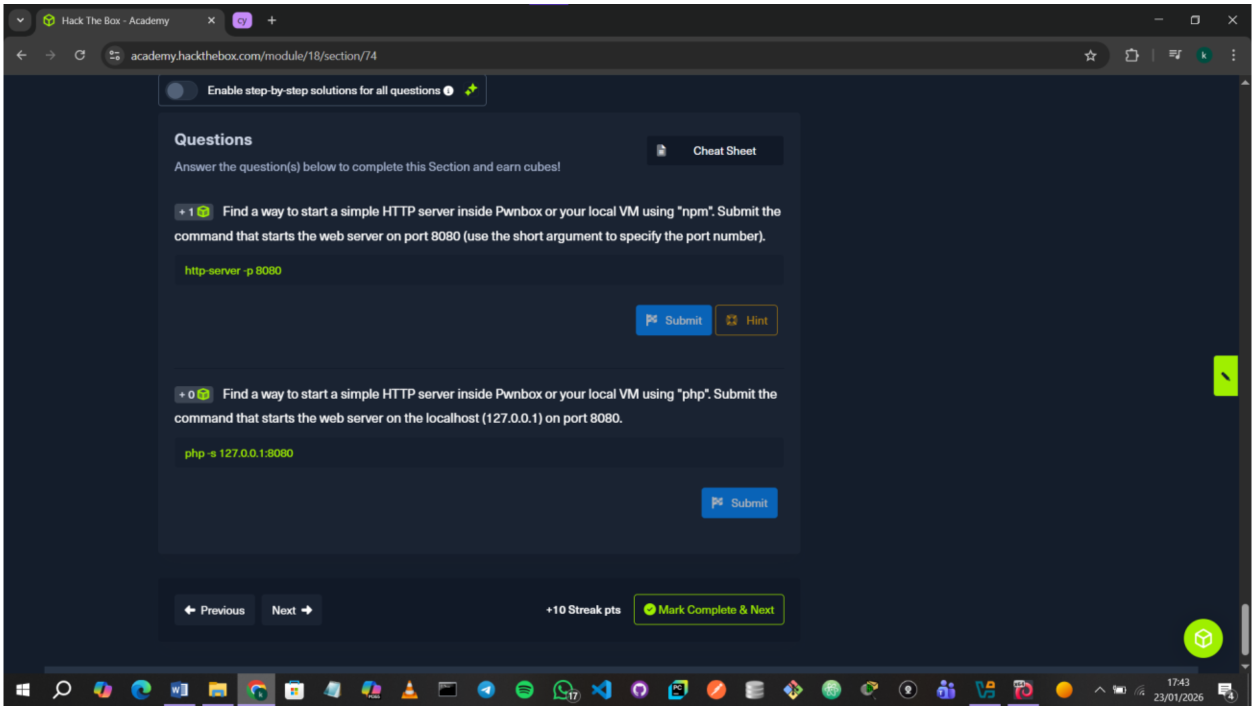
Answer1: I first located it by running **sudo find / -name ‘dconf.service’ 2>/dev/null**. I then ran **cat /usr/lib/system/user/dconf.service** to find the Type of service which was “dbus”.

4.5 Network Services

In this section, I learned about common Linux network services and their roles in system administration and security. I understood how **SSH** is used for secure remote access and why encryption is important when connecting to remote systems. I also learned about **NFS** and how it allows files to be shared across a network as if they were stored locally, including the importance of correctly setting permissions. Additionally, I learned how **web servers** are used to serve files and applications over HTTP and how they can also be used for simple file transfers. Finally, I learned how **VPNs** create encrypted tunnels that allow secure access to internal networks from remote locations.

4.6 Working with Web Services

In this module, I learned how web servers enable communication between browsers and servers. I focused on **Apache**, which acts like the engine of a website, handling requests and serving content. Its modularity allows it to be extended with features like **mod_ssl** for encrypted communication, **mod_proxy** for routing requests, and modules to manipulate headers and URLs.



4.7 Backup and Restore

In this module, I learned how Linux systems handle data backup and restoration to prevent data loss. The module emphasized the importance of **encryption** in backups to prevent unauthorized access to sensitive data. I learned that tools such as SSH, GnuPG, and disk encryption technologies can be used to ensure data confidentiality during storage and transfer.

4.8 File System Management

In this section, I learned how Linux manages file systems, storage devices, and disk partitions. I understood the differences between common file systems such as ext4, Btrfs, XFS, and NTFS, and how choosing a file system depends on factors like performance, reliability, and compatibility. I also learned how the Linux file system is structured hierarchically and how inodes store metadata that helps the system track files efficiently.

Question1: How many partitions exist in our Pwnbox?

Answer1: I only found one but according to the module there were 3.

4.9 Containerization

In this module, I learned the basics of containerization and how containers run applications in isolated, lightweight environments. I understood the difference between containers and virtual machines and why containers are more efficient. I learned how Docker and LXC are used to create, run, and manage containers. I also learned about container security, resource isolation, and why proper configuration is important.

5.0 Linux Networking

5.1 Network Configuration

In this section, I learned how to configure and manage network settings on Linux systems. I learned how to view and configure network interfaces, assign IP addresses, set gateways, and configure DNS. I also learned about network access control models (DAC, MAC, and RBAC), basic network monitoring, troubleshooting tools, and Linux hardening mechanisms such as SELinux, AppArmor, and TCP wrappers.

5.2 Remote Desktop Protocols in Linux

In this section, I learned about remote desktop protocols used in Linux for graphical remote access. I learned how protocols like RDP and VNC allow administrators to manage systems remotely, and how X11 enables network-transparent graphical sessions. I also learned about X11 forwarding over SSH, common ports used by these services, and the security risks associated with insecure protocols like X11 and XDMCP, as well as why VNC is more commonly and securely used in Linux environments.

6.0 Linux Hardening

6.1 Linux Security

In this section, I learned the core security fundamentals for securing Linux systems. I learned the importance of keeping systems updated, using firewalls, securing SSH access, and applying the principle of least privilege. I also learned about tools and mechanisms such as fail2ban, SELinux/AppArmor, system auditing, and TCP wrappers, and how they help restrict access, reduce attack surfaces, and prevent unauthorized access to Linux services.

6.2 Firewall Setup

In this section, I learned how firewalls are used in Linux to control and secure network traffic. I learned about the Netfilter framework and how iptables is used to define firewall rules using tables, chains, rules, matches, and targets. I also learned the purpose of common tables like filter and nat, how traffic is handled through chains such as INPUT and OUTPUT, and how rules are created to allow, block, or modify network traffic based on specific conditions.

6.3 System Logs and Monitoring

In this section, I learned how system logs in Linux are used to monitor system activity, troubleshoot issues, and detect security threats. I learned about different log types such as kernel, system, authentication, application, and security logs, and where they are stored. I also learned how logs can be analyzed to identify failed logins, suspicious activity, misconfigurations, and potential attacks.

7.0 Linux Distribution vs Solaris

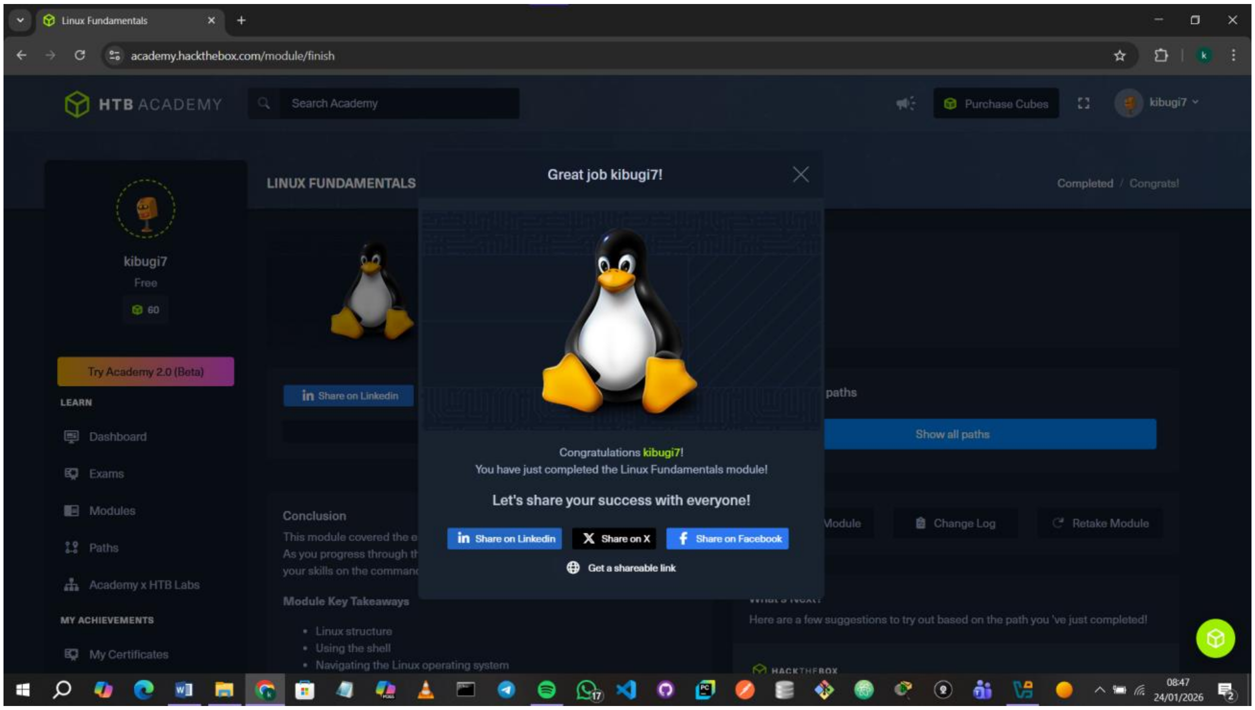
7.1 Solaris

In this section, I learned that Solaris is a Unix-based, enterprise-focused operating system designed for high performance, stability, and security. I learned how it differs from Linux in areas such as package management, process monitoring, permissions, NFS configuration, and system management.

8.0 Tips & Tricks

8.1 Shortcuts

In this part, I learned several Linux keyboard shortcuts that make working in the terminal faster and more efficient.



9.0 Conclusion

The Hack The Box Linux Fundamentals module helped me understand how Linux works and how to interact with it confidently through the terminal. I learned essential commands, file and permission management, and productivity shortcuts that improved my efficiency. I had to do a lot of research for the questions but it is all a part of the learning process I guess. Overall, this module strengthened my Linux skills and prepared me for more advanced cybersecurity tasks.